

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56d)

Metric

$$ds^2 = \left(-\frac{\kappa_0^2 q_0^4 e^x + 4(a\kappa_0 e^x - \kappa_0 m_0)q_0^2 r + 2a^2 r u}{2\kappa_0 q_0^2 r^2} \right) du \otimes du - du \otimes dr - dr \otimes du + r^2 e^x dx \otimes dx + r^2 e^x dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (-1) du$$

$$n = \left(-\frac{\kappa_0 q_0^2 e^x}{4r^2} - \frac{ae^x}{r} + \frac{m_0}{r} - \frac{a^2 u}{2\kappa_0 q_0^2 r} \right) du + (-1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r e^{(\frac{1}{2} x)} \right) dx + \left(\frac{1}{2} i \sqrt{2} r e^{(\frac{1}{2} x)} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r e^{(\frac{1}{2} x)} \right) dx + \left(-\frac{1}{2} i \sqrt{2} r e^{(\frac{1}{2} x)} \right) dy$$

Contravariant components

$$l = (1) dr$$

$$n = (1) du + \left(-\frac{\kappa_0 q_0^2 e^x}{4r^2} - \frac{ae^x}{r} + \frac{m_0}{r} - \frac{a^2 u}{2\kappa_0 q_0^2 r} \right) dr$$

$$m = \left(\frac{\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \right) dx + \left(\frac{i\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \right) dx + \left(-\frac{i\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \right) dy$$

Directional Derivatives

$$D = \partial_r$$

$$\Delta = \partial_u - \frac{\kappa_0 q_0^2 e^x}{4r^2} - \frac{ae^x}{r} + \frac{m_0}{r} - \frac{a^2 u}{2\kappa_0 q_0^2 r} \partial_r$$

$$\delta = \frac{\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \partial_x + \frac{i\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \partial_x - \frac{i\sqrt{2}e^{(-\frac{1}{2}x)}}{2r} \partial_y$$

Spin Coefficients

$$\rho = -\frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\kappa_0 q_0^2 e^x}{4r^3} - \frac{ae^x}{r^2} + \frac{m_0}{r^2} - \frac{a^2 u}{2\kappa_0 q_0^2 r^2}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}\kappa_0 q_0^2 e^{\left(\frac{1}{2}x\right)}}{8r^3} + \frac{\sqrt{2}ae^{\left(\frac{1}{2}x\right)}}{2r^2}$$

$$\alpha = -\frac{\sqrt{2}e^{\left(-\frac{1}{2}x\right)}}{8r}$$

$$\beta = \frac{\sqrt{2}e^{\left(-\frac{1}{2}x\right)}}{8r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{\kappa_0 q_0^2 e^x}{4r^3} - \frac{ae^x}{2r^2} + \frac{m_0}{2r^2} - \frac{a^2 u}{4\kappa_0 q_0^2 r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\kappa_0 q_0^2 e^x}{4 r^4}$$

$$\Phi_{12} = -\frac{\sqrt{2}\kappa_0 q_0^2 e^{\left(\frac{1}{2} x\right)}}{8 r^4} - \frac{\sqrt{2} a e^{\left(\frac{1}{2} x\right)}}{4 r^3}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{\sqrt{2}\kappa_0 q_0^2 e^{\left(\frac{1}{2} x\right)}}{8 r^4} - \frac{\sqrt{2} a e^{\left(\frac{1}{2} x\right)}}{4 r^3}$$

$$\Phi_{22} = \frac{\kappa_0 q_0^2}{8 r^4} + \frac{a}{2 r^3} + \frac{a^2}{2 \kappa_0 q_0^2 r^2}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{\kappa_0 q_0^2 e^x}{2 r^4} + \frac{a e^x}{r^3} - \frac{m_0}{r^3} + \frac{a^2 u}{2 \kappa_0 q_0^2 r^3}$$

$$\Psi_3 = -\frac{\sqrt{2}\kappa_0 q_0^2 e^{\left(\frac{1}{2} x\right)}}{4 r^4} - \frac{3 \sqrt{2} a e^{\left(\frac{1}{2} x\right)}}{4 r^3}$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"